

RESEARCH ENGINEER · LUMA AI

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Research Engineer at Luma AI working on *structured visual generation* — models that turn flat pixels into editable raster, text, and vector components for humans and agents. I lead *Layering*, built the Ray3 RL workflow, and have published work on multimodal evaluation and 4D grounding.

Stanford **MSCS**, *Distinction in Research*, advised by Jiajun Wu. 1 first-author **NeurIPS 2024**, co-first **CVPR 2026** (accepted), contributor **NeurIPS 2025**.
Edinburgh **BEng** joint Honors, ranked **2nd** of cohort.

I — EDUCATION

Education.

2019 — 2025

M.S. Computer Science	Stanford University M.S. Computer Science GPA 3.97 / 4.00 · Distinction in Research Advised by Prof. Jiajun Wu — led 1 first-author (IKEA Video Manuals, NeurIPS '24) and 2 collaborative projects on 4D scene grounding, multimodal evaluation, and probing emergent structure in video-diffusion models.	Sep 2023 — Apr 2025 Stanford, CA
BEng Electronics & CS	The University of Edinburgh BEng Electronics & Computer Science · Joint Honors · Ranked 2nd of cohort	Sep 2019 — Jun 2023 Edinburgh, UK
Visiting ECE	The University of Texas at Austin Electrical & Computer Engineering · 18 credits / semester	Jan — Jun 2022 Austin, TX

II — EXPERIENCE

Research *experience*.

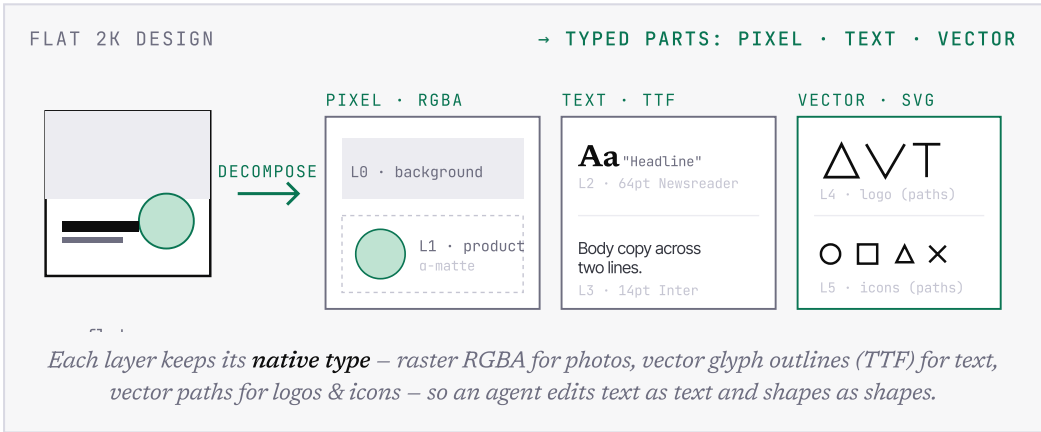
2023 — present

2025 April — present
Palo Alto, CA

Luma AI · Research Engineer (Project Lead)

Generative models · multimodal · agentic systems

- Layering** — project lead. Lead Layering research at Luma. Turn flat image generations into *ordered, editable RGBA layer stacks with per-layer captions*, so agents and humans can extract, add, decompose & remix elements without regenerating. End-to-end ownership across data, model, eval, and agent loop.



- Layering** — systems. **Data**: PSD / Canva / Crello / synthetic pipelines (>1.5M layered samples). **Model**: two-tower RGBA tokenizer, layer-aware diffusion, decoder fine-tuning at 768–2K. **Eval**: native-res harness, layer-level autocaliper, frozen 2K test sets. **Agent loop** that inspects the canvas, proposes edits, and validates the result against user intent.
- Ray3 (video generation)**. Built the Ray3 **RL workflow** and ran reward, data, and evaluation experiments for video generation quality and prompt following.
- Uni-1 (unified multimodal foundation model)**. Ran **RL** and other experiments, along with data work around unified multimodal understanding and generation.
- Reward modeling & calibration**. Built reward and calibration experiments for visual generators — prompt following, composition, aesthetics — calibrated against held-out human preference and designed to resist reward over-optimization.
- Cross-team leadership**. Run weekly research syncs and a recurring tech-artist sync on synthetic data; onboarded new researchers; coordinate with infra and product teams on integration.

- **IKEA Video Manuals — NeurIPS 2024 (first author)**. Sole student lead on a year-long project: first dataset that aligns real-world assembly videos with 3D models in space and time. Built the cross-frame optimization combining PnP-RANSAC with temporal-consistency constraints; coordinated 30 annotators across 34k+ frames over 98 videos with iterative cross-validation; defined the evaluation protocol now used by follow-up 4D-grounding work.
- **Zero-shot optical flow from video diffusion — NeurIPS 2025 (contributor)**. Helped develop a counterfactual probe over video-diffusion logits that yields state-of-the-art TAP-Vid optical flow with no labels and no fine-tuning; generalizes to in-the-wild videos and outperforms specialized baselines (*Taming generative video models for zero-shot optical flow extraction*, S. Kim*, K. L. Aw*, K. Kotar* et al.).
- **Distinction in Research** · 4 quarters of funded RAship.

Selected *publications*.

2026

CVPR

CaptionQA — Is your caption as useful as the image itself?

S. Yang*, **Y. Liu***, B. Zhai*, X. Sun, Z. Liu, E. Barsoum, M. Li, C. Xu

Utility-based caption benchmark — 33k MCQs across Natural / Document / E-commerce / Embodied; surfaces image-caption utility gaps standard QA-on-image benchmarks miss.

arXiv 2511.21025 github / bronyayang → [paper](#)

2025

NeurIPS

Taming generative video models for *zero-shot optical flow* extraction

S. Kim*, K. L. Aw*, K. Kotar*, C. Eyzaguirre, W. Lee, **Y. Liu**, J. Watrous, S. Stojanov, J. C. Niebles, J. Wu, D. L. K. Yamins

Counterfactual probe over video-diffusion logits — SOTA TAP-Vid optical flow with no labels and no fine-tuning.

arXiv 2507.09082 code / kl_tracing → [project page](#)

2024

NeurIPS
Datasets &
Benchmarks Track

IKEA Manuals at Work — 4D grounding of assembly instructions on internet videos

Y. Liu, C. Eyzaguirre, M. Li, S. Khanna, J. C. Niebles, V. Ravi, S. Mishra, W. Liu*, J. Wu*

First dataset aligning real-world assembly videos with 3D models — 34k+ frames, 98 videos. Sole student lead, year-long project; coordinated 30-annotator cross-validation pipeline.

arXiv 2411.11409 page / ikea-video → [project page](#)

2022

JMIR

Infodem. **CO-FIRST**

COVID-19 misinformation detection — machine-learned solutions to the *infodemic*

Y. Liu*, N. Kolluri*, D. Murthy

Hybrid classical + pre-trained LM framework with large-scale crowdsourced annotation; shows low-agreement crowd labels can still meaningfully improve performance.

JMIR Infodem. 2(2) · e38756 → [paper](#)

What I'm *building recently*.

Interactive visual generation.

LUMA AI · LAYERING · PROJECT LEAD

Today's image and video generators are fundamentally *turn-based*: a prompt goes in, a flat raster comes out, and the only way to change anything is to re-roll the whole thing. But real creative work isn't a single turn — it's a conversation. People want to *point, edit, redirect, and revise* mid-stream; agents want to read the canvas, propose a change, and verify the result. Neither is possible when the output is an opaque pixel grid.

My current interest is making generative visual models *interactive by design* — bridging pixel outputs with code-backed representations so humans and agents can both stay in the loop. **Layering** is my current effort in this direction: the model emits an ordered *RGBA stack* with per-layer captions instead of a flat raster — a representation that pixel-domain models and agents with code-backed tools can both operate on. From there, editing becomes a first-class operation, and a long-horizon agent can revise the canvas and verify its own outputs against the user's intent.

RESEARCH INTERESTS

I'm interested in generative systems whose outputs are inspectable, editable, and verifiable — especially where pixel outputs connect to structured or code-backed representations.

REWARD	RLHF and reward modeling for flow-matching generators; calibration against held-out human preference; reward models that resist over-optimization.
STRUCTURED OUTPUT	Generation as structured, multi-layer output that agents can read and revise — instead of opaque rasters that can only be re-rolled.
EVAL	Utility-based multimodal benchmarks (CaptionQA, IKEA Manuals) that measure whether outputs are actually useful to humans and downstream agents.